Information Systems Management

Chapter # 5

IT Infrastructure and Emerging Technologies

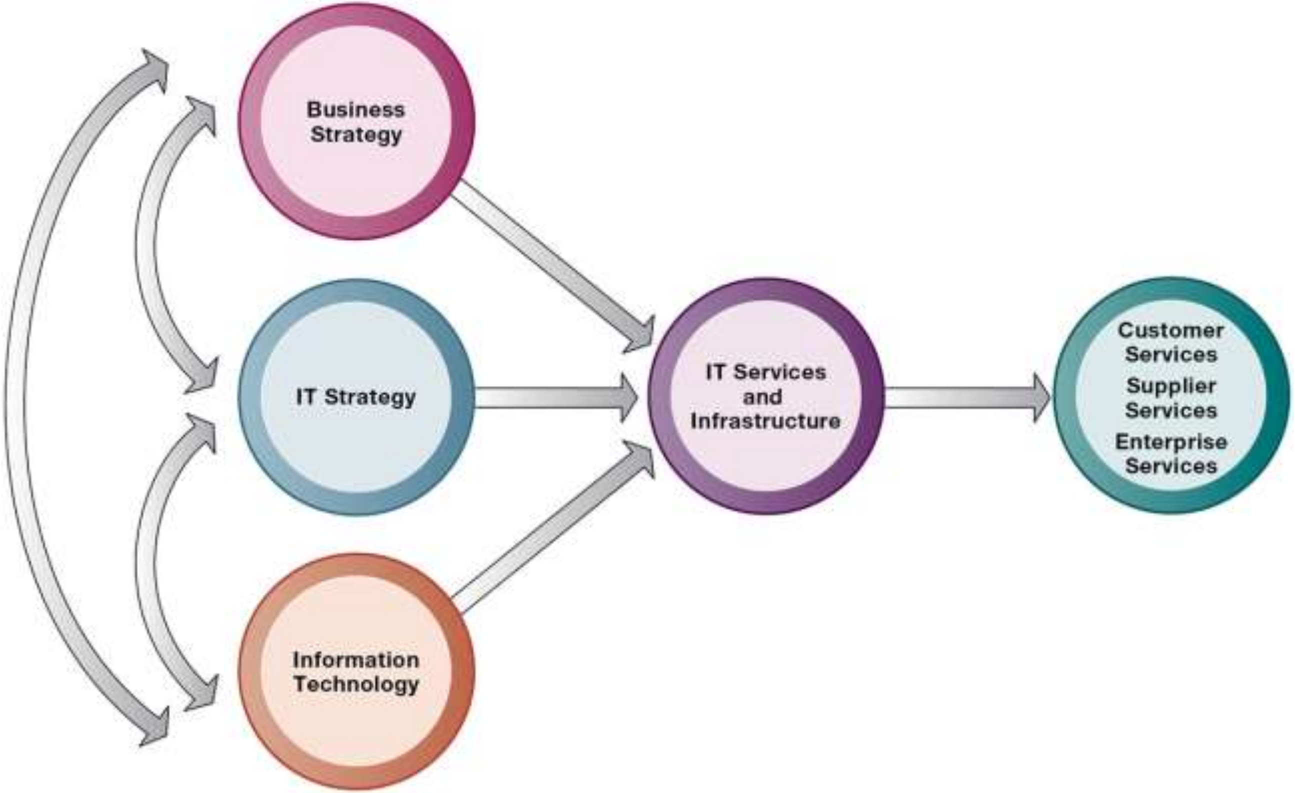
The slides have been prepared using the above text book.

Pearson

Case Study Presentation

#1 Grab leverages IT to enhance its services

Connection Between the Firm, IT Infrastructure, and Business Capabilities



Defining IT Infrastructure

- Set of physical devices and software required to operate the entire enterprise.
- Set of firm-wide services including:
    - Computing platforms providing computing services
    - Physical facilities management services
    - I T management, education, and other services
- “Service platform” perspective
    - Refers to analyzing the actual services enabled by new technology tools
    - More accurate view of value of investments

Defining IT Infrastructure

- Computing Services
- Telecommunication Services
- Data Management Services
- Application Software Services
- Physical Facilities Management Services
- IT Management Services
- IT Standards Services
- IT Education Services
- IT Research and Development Services

Evolution of IT Infrastructure

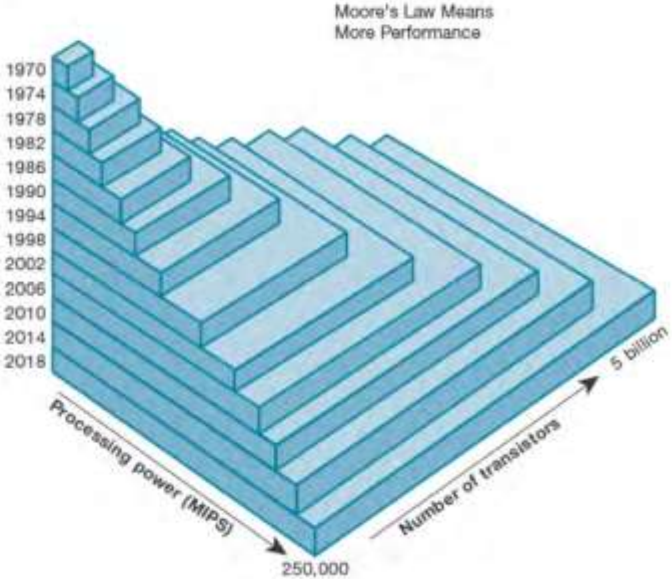
- 1959 to present : General-purpose mainframe and minicomputer era
- 1981 to present : Personal computer era
- 1983 to present : Client/server era
- 1992 to present : Enterprise computing era
- 2000 to present : Cloud and mobile computing era
- 2010 to Present : Big Data & Deep Learning era
- 2020 to Present : Generative AI and Foundation Models era

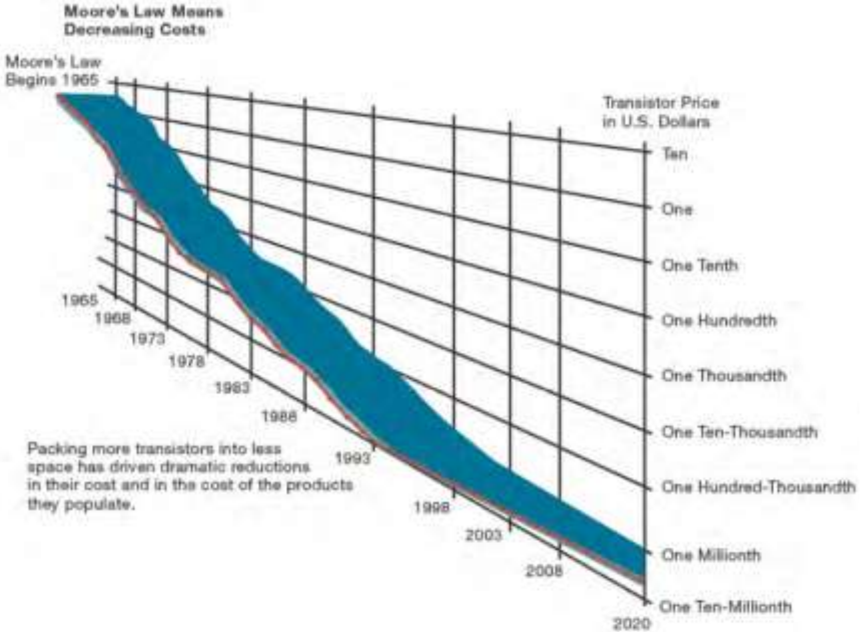
Technology drivers of Infrastructure Evolution

- Moore's law and Microprocessing Power
  - # of transistors will double each year
  - Nanotechnology
- Law of Mass Digital Storage
  - The amount of data being stored each year doubles
- Metcalfe's Law and network economics
  - Value or power of a network grows exponentially as a function of the number of network members

Technology drivers | Moore’s Law

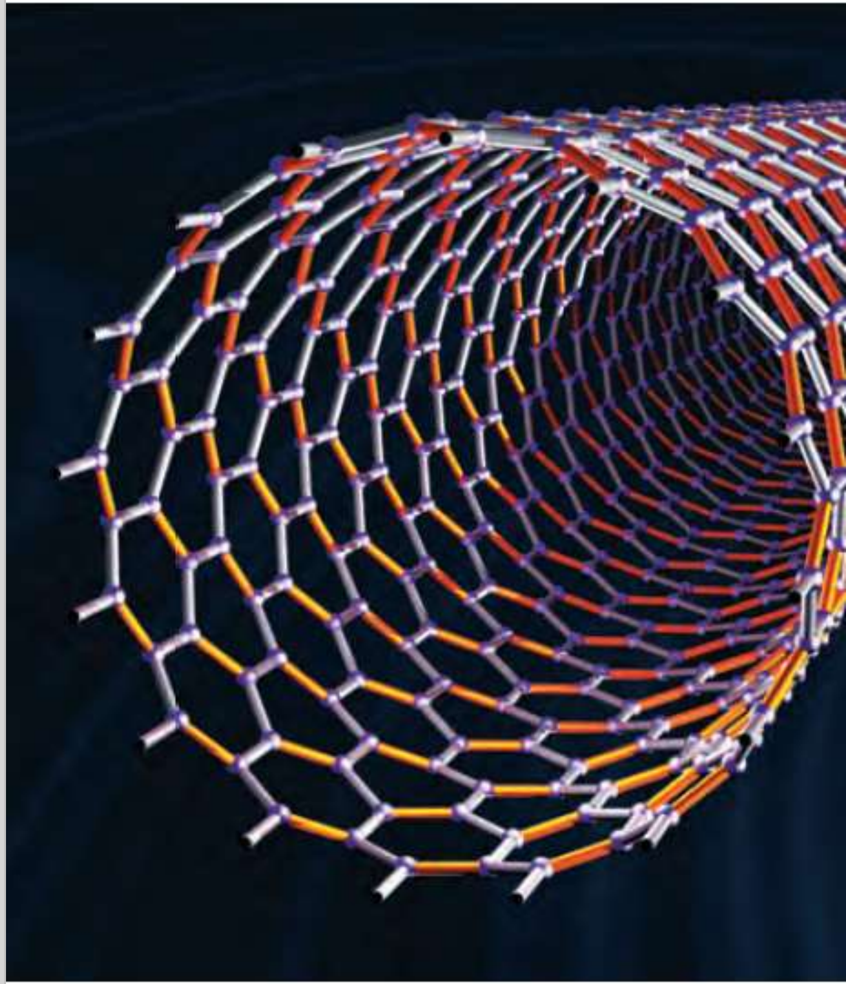
<u>Variants of Moore’s Law </u>: (1) the power of microprocessors doubles every 18 months, (2) computing power doubles every 18 months, and (3) the price of computing falls by half every 18 months.

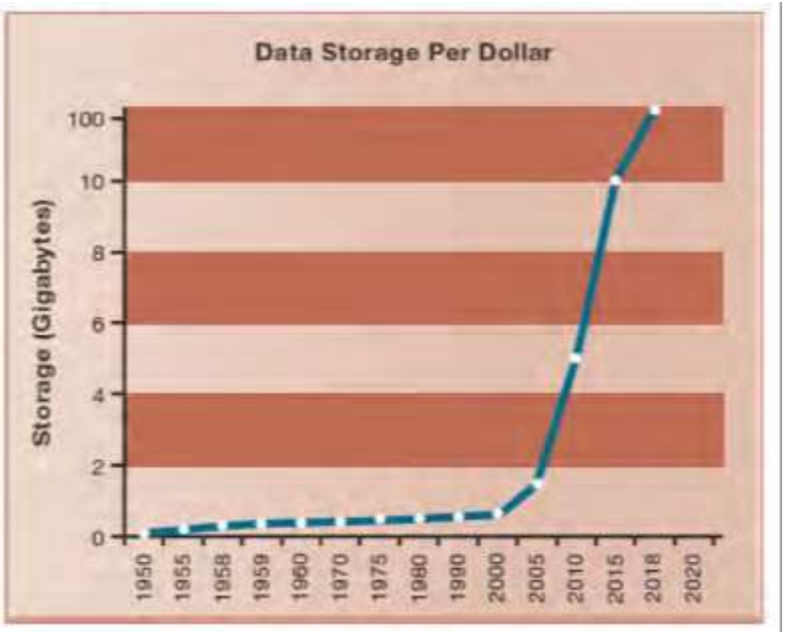




Technology drivers | Nanotechnology

- **Nanotechnology** uses individual atoms and molecules to create computer chips and other devices that are thousands of times smaller than current technologies permit.
- Chip manufacturers can shrink the size of transistors down to the width of several atoms by using nanotechnology.
- Chip manufacturers are trying to develop a manufacturing process to produce nanotube processors economically.
- Stanford University scientists have built a nanotube computer.





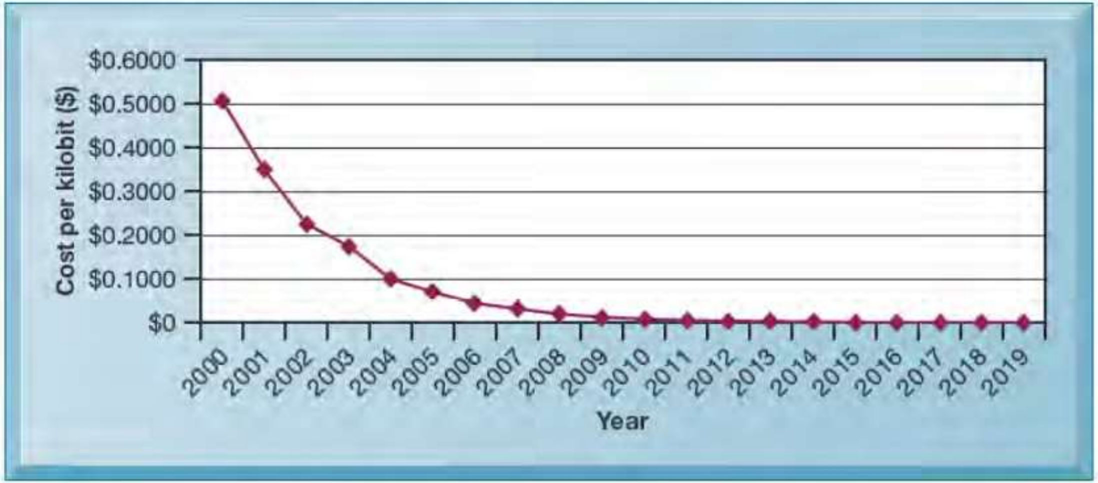
Technology drivers | Law of Mass digital Storage

- The amount of digital information is roughly doubling every year (Lyman and Varian, 2003).
- The cost of storing digital information is falling at an exponential rate of 100 percent a year
- Google provides 100GB of data at $1.99 per month

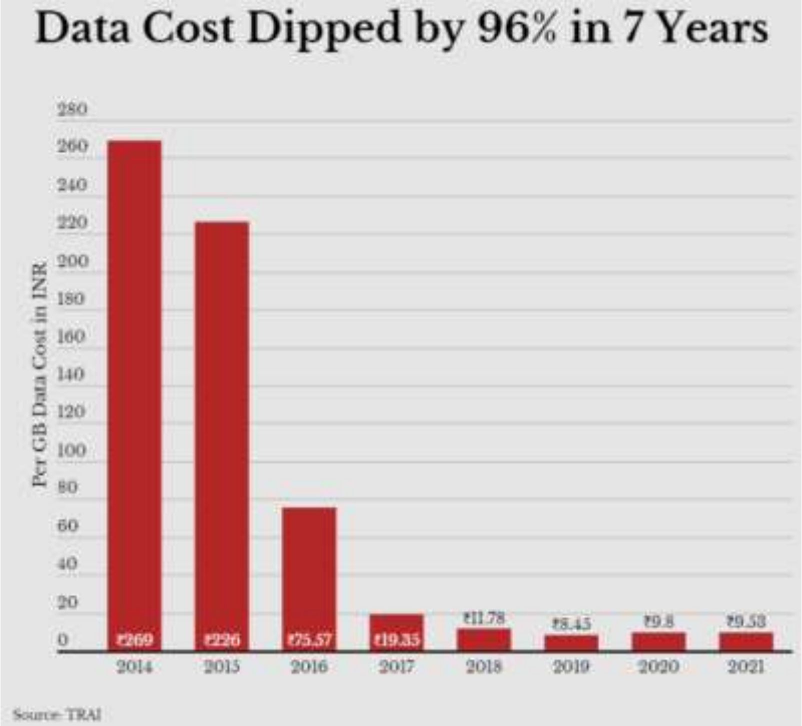
Technology drivers | Metcalfe's Law

- Robert Metcalfe, inventor of Ethernet local area network technology
- The value or power of a network grows exponentially as a function of the number of network members (1970)
- Point to the increasing returns to scale that network members receive as more and more people join the network.

Technology drivers | Declining Communications Cost & Internet



US - Avg Internet Connection Speed

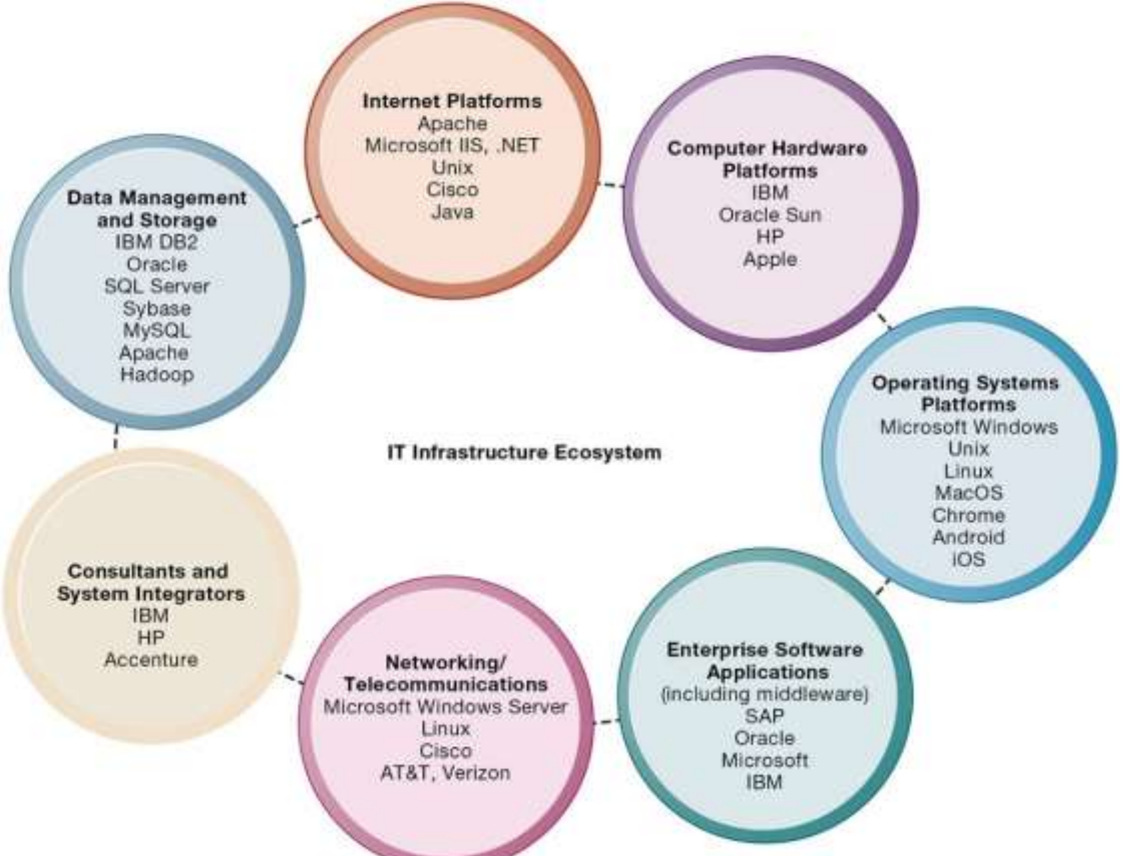


India

Technology drivers | Acceptance of Technology Standards

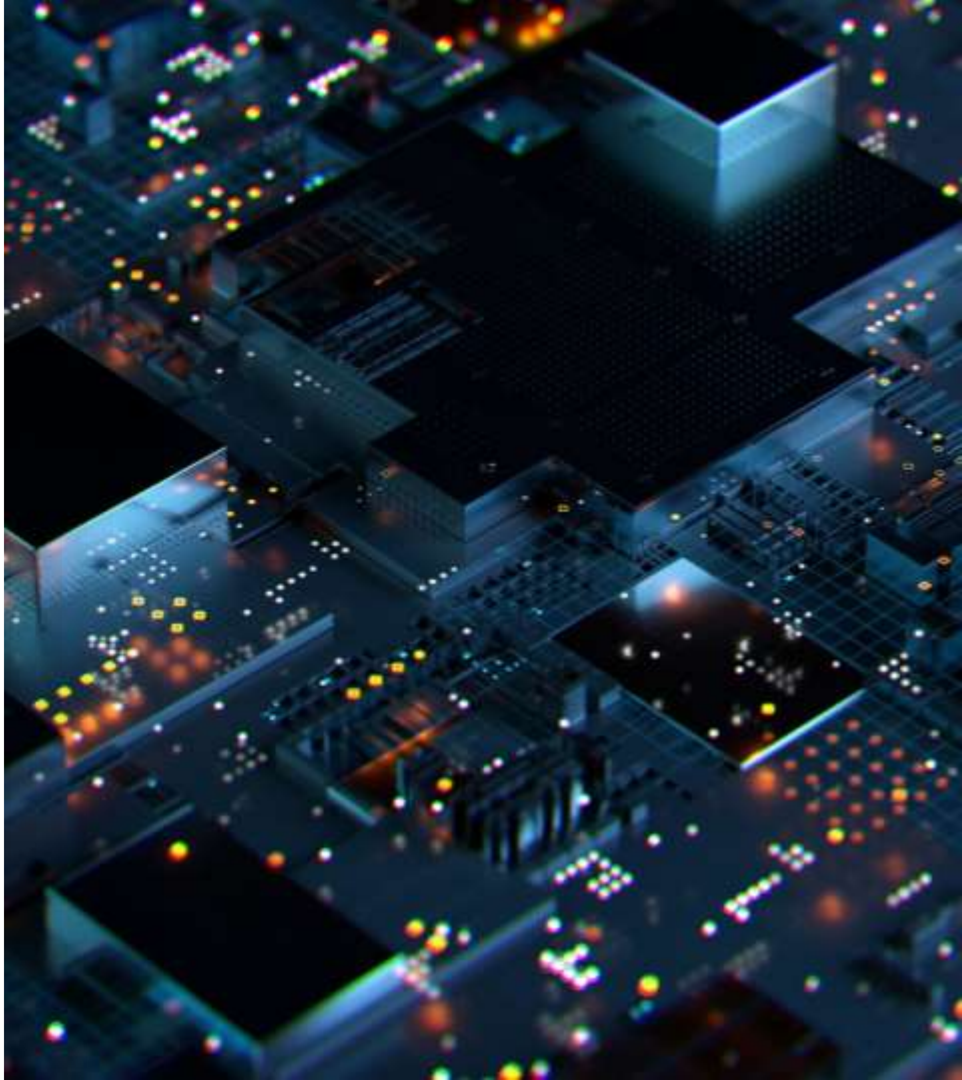
STANDARD	SIGNIFICANCE
American Standard Code for Information Interchange (ASCII) (1958)	Made it possible for computer machines from different manufacturers to exchange data; later used as the universal language linking input and output devices such as keyboards and mice to computers. Adopted by the American National Standards Institute in 1963.
Common Business Oriented Language (COBOL) (1959)	An easy-to-use software language that greatly expanded the ability of programmers to write business-related programs and reduced the cost of software. Sponsored by the Defense Department in 1959.
Unix (1969–1975)	A powerful multitasking, multiuser, portable operating system initially developed at Bell Labs (1969) and later released for use by others (1975). It operates on a wide variety of computers from different manufacturers. Adopted by Sun, IBM, HP, and others in the 1980s, it became the most widely used enterprise-level operating system.
Ethernet (1973)	A network standard for connecting desktop computers into local area networks that enabled the widespread adoption of client/server computing and local area networks and further stimulated the adoption of personal computers.
Transmission Control Protocol/Internet Protocol (TCP/IP) (1974)	Suite of communications protocols and a common addressing scheme that enables millions of computers to connect together in one giant global network (the Internet). Later, it was used as the default networking protocol suite for local area networks and intranets. Developed in the early 1970s for the U.S. Department of Defense.
IBM/Microsoft/Intel Personal Computer (1981)	The standard Wintel design for personal desktop computing based on standard Intel processors and other standard devices, Microsoft DOS, and later Windows software. The emergence of this standard, low-cost product laid the foundation for a 25-year period of explosive growth in computing throughout all organizations around the globe. Today, more than 1 billion PCs power business and government activities every day.
World Wide Web (1989–1993)	Standards for storing, retrieving, formatting, and displaying information as a worldwide web of electronic pages incorporating text, graphics, audio, and video enable creation of a global repository of billions of web pages.

Components of IT Infrastructure



Computer Hardware Platforms

- Client machines
  - Desktop PCs, laptops
  - Mobile computing: smartphones, tablets
  - Desktop chips vs. mobile chips
- ServersAnd telecommunications networks



Operating System Platforms

- Corporate servers
  - Windows Server
  - Unix
  - Linux

- Client level
  - Microsoft Windows
  - Android, iOS, Windows 10+ (mobile/multitouch)
  - Google’s Chrome OS(cloud computing)

Enterprise Software Applications

- In 2020, firms are expected to spend over $500 billion on software for enterprise applications.
- These applications are considered to be part of IT infrastructure. Traditionally, large firms were the most prominent users of enterprise software, but firms such as Microsoft have moved into the market of small- and medium-size businesses that can benefit from enterprise software.
- Some of the Largest providers: SAP and Oracle
- Some of the largest Middleware providers: IBM, Oracle

Database Management Systems

- Database software providers
  - IBM (DB2)
  - Oracle
  - Microsoft (SQL Server)
  - SAP Sybase (Adaptive Server Enterprise),
  - MySQL (Oracle)
  - Apache Hadoop
  - Etc etc

Networking/Telecommunications

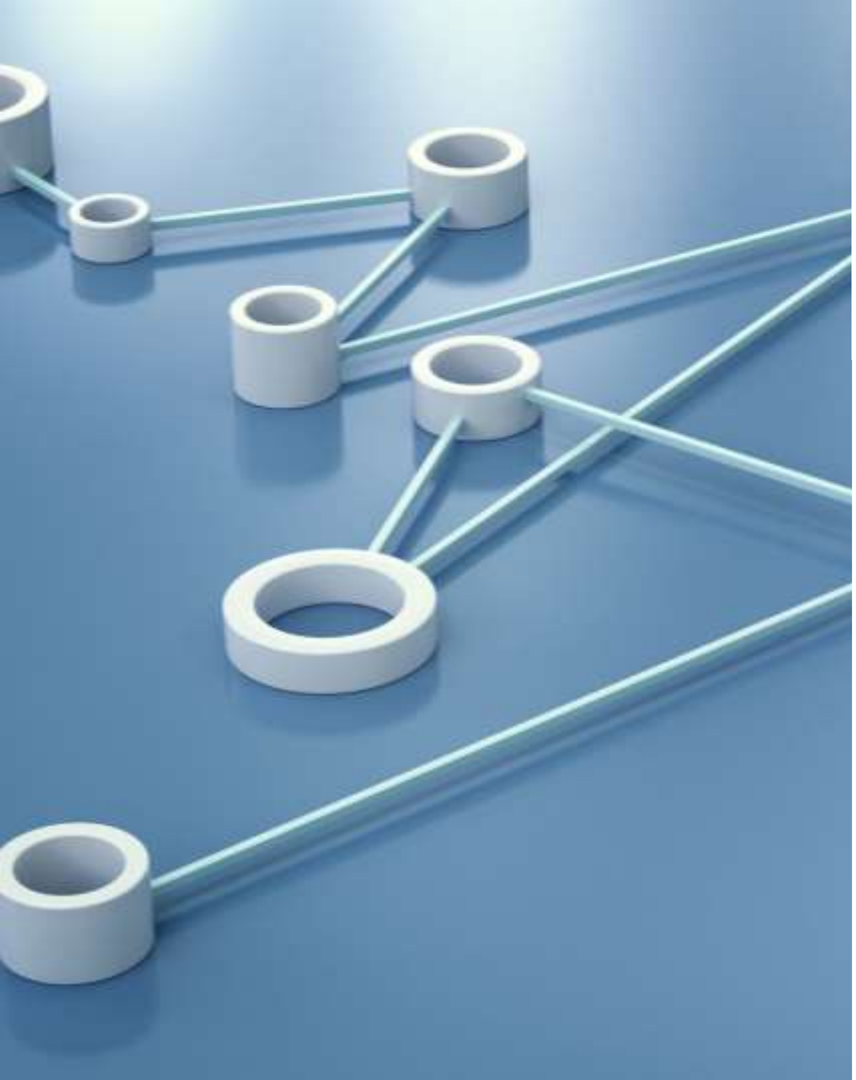
- Windows Server is predominantly used as a local area network operating system, followed by Linux and Unix.
- Most local area networks, as well as wide area enterprise networks, use the TCP/IP protocol suite as a standard .
- Cisco and Juniper Networks are leading networking hardware providers.
- Telecommunication Services – voice and data connectivity, wide area networking, wireless services, and Internet access. Leading Vendors – AT&T, Verizon, (in India – Airtel, Jio, Tata Communications etc)

Internet Platforms

- Includes hardware, software, and management services to support a firm's website, including web hosting services, routers, and cabling or wireless equipment.

- **DNS hosting** is a service that manages your domain's records to find your website, ensuring users are directed to the correct server. Eg: Cloudflare, CloudDNS, Amazon Route 53, Google Cloud DNS etc

- A **web hosting service** maintains a large web server, or series of servers, and provides fee-paying subscribers with space to maintain their websites.



Consulting and System Integration Services

- Even large firms do not have resources for full range of support for new, complex infrastructure
- Leading consulting firms: Accenture, IBM Global Services, HP, Infosys, Wipro Technologies
- Software integration: ensuring new infrastructure works with legacy systems
- Legacy systems: older TPS created for mainframes that would be too costly to replace or redesign

Trends in Computer Hardware Platforms (1/5)

- The **mobile digital** platform
  - Smartphones
  - Tablet computers
  - Digital e-book readers and apps (Kindle)
  - Wearable devices

- **Consumerization** of IT and BYOD (bring your own device)
  - Forces businesses and IT departments to rethink how IT equipment and services are acquired and managed

Trends in Computer Hardware Platforms (2/5)

- **Quantum computing**
  - Uses quantum physics to represent and operate on data (IBM)
  - Dramatic increases in computing speed and power.

- **Virtualization**
  - Allows single physical resource to act as multiple resources (i.e., run multiple instances of OS); also enables multiple physical resources (such as storage devices) to appear as a single logical resource (such as in software-defined storage (SDS))(VMWare)
  - Reduces hardware and power expenditures
  - Facilitates hardware centralization

Trends in Computer Hardware Platforms (3/5)

- **Cloud computing**
  - On-demand computing services obtained over network
    - Infrastructure as a service (IaaS)
    - Software as a service (SaaS)
    - Platform as a service (PaaS)
  - Cloud can be public or private
  - Allows companies to minimize IT investments
  - Drawbacks: Concerns of security, reliability
  - Hybrid cloud computing model

Trends in Computer Hardware Platforms (4/5)

- **Edge computing**
  - Servers at the edge of the network, near the source of the data
  - Reduces latency and network traffic
  - Edge computing is needed to provide efficient service to consumers located across the country.
  - It works by reducing the distance between local computers and web resources and sharing the total load across many regional computing facilities.

Trends in Computer Hardware Platforms (5/5)

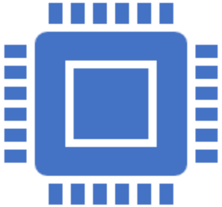
Green computing (Green IT)

Data centers consume ~1.5% of all energy use in the world.

Practices and technologies for manufacturing, using, disposing of computing and networking hardware

Reducing power consumption a high priority

Green data centers



High-performance, power-saving processors

Multicore processors

Power-efficient microprocessors

Trends in Computer Software Platforms (1/3)

Linux and open-source software

Produced by community of programmers

Examples: Apache web server, Mozilla Firefox browser, OpenOffice

Linux

Software for the web: Java, H T M L, and H T M L 5

Java Virtual Machine

Web browsers

H T M L and H T M L 5

Ruby and Python

Trends in Computer Software Platforms (2/3)

Web services and service-oriented architecture

- Web services
- XML : Extensible Markup Language
- SOA : service-oriented architecture
    - Set of self-contained services that communicate with one another to create a working software application
    - Software developers reuse these services in other combinations to assemble other applications as needed

Trends in Computer Software Platforms (3/3)

- Software Outsourcing and Cloud Services
  - Software Packages & Enterprise Software
  - Software Outsourcing
  - Cloud based software services and tools
  - Mashups and Apps

Challenges in managing IT Infrastructure (1/3)

- Dealing with platform and Infrastructure change
  - How to remain flexible ? Scalable ?
  - Cloud computing ? New policies ?
  - BYOD ? MDM ?

- Dealing with platform and Infrastructure change
  - Refer to earlier sessions
  - Should departments have responsibility of IT decisions or should IT be centrally controlled ?

Challenges in managing IT Infrastructure (2/3)

- TCO : Making wise Infrastructure Investments
- TCO = Total cost of ownership of technology assets

INFRASTRUCTURE COMPONENT	COST COMPONENTS
Hardware acquisition	Purchase price of computer hardware equipment, including computers, terminals, storage, and printers
Software acquisition	Purchase or license of software for each user
Installation	Cost to install computers and software
Training	Cost to provide training for information systems specialists and end users
Support	Cost to provide ongoing technical support, help desks, and so forth
Maintenance	Cost to upgrade the hardware and software
Infrastructure	Cost to acquire, maintain, and support related infrastructure, such as networks and specialized equipment (including storage backup units)
Downtime	Cost of lost productivity if hardware or software failures cause the system to be unavailable for processing and user tasks
Space and energy	Real estate and utility costs for housing and providing power for the technology

Challenges in managing IT Infrastructure (3/3)

Competitive Forces model for IT Infrastructure